An Improved Analysis of the Empirical Convergence Rate of DPO-Mix-R

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A Dimension-Free Moment-Doubling Analysis

We present a moment-based refinement of the empirical convergence analysis.

Let

$$\delta_t(a,a') := \delta(a,a';\theta^{(t)}), \qquad a,a' \in \mathcal{Y}.$$

For $p \geq 2$, define

$$M_t(p) := \max_{a,a' \in \mathcal{Y}} \left(\mathbb{E}|\delta_t(a,a')|^p\right)^{1/p}.$$

The empirical update can be decomposed as

$$\delta_{t+1}(a,a') = \delta_{t+1}^{\mathrm{ex}}(a,a') + \nu_t(a,a'),$$

where $\delta_{t+1}^{\mathrm{ex}}$ is the one-step exact-gradient image from the current empirical iterate, and ν_t is the centered empirical-gradient noise.

For DPO-Mix-R, the exact-gradient part satisfies

$$|\delta_{t+1}^{\mathrm{ex}}(a,a')| \leq \frac{1}{4A} \sum_{b \in \mathcal{Y}} \left(\delta_t(a,b)^2 + \delta_t(a',b)^2\right). \tag{1}$$

Equivalently, this is the averaged quadratic bound with the universal constant $K = 1/2$.

We also use the following consequence of the empirical sub-Gaussian noise assumption: for every $p \geq 2$,

$$\|\nu_t(a,a')\|_{L^p} \leq 3\sigma\sqrt{p}, \qquad \forall a,a' \in \mathcal{Y},\ t \geq 0. \tag{2}$$

Lemma 1 (Moment-doubling recursion)**.** *For every $p \geq 2$ and every $t \geq 0$,*

$$M_{t+1}(p) \leq \frac{1}{2} M_t(2p)^2 + 3\sigma\sqrt{p}.$$

Proof. Fix $a,a' \in \mathcal{Y}$. By the empirical decomposition and the triangle inequality,

$$\|\delta_{t+1}(a,a')\|_{L^p} \leq \|\delta_{t+1}^{\mathrm{ex}}(a,a')\|_{L^p} + \|\nu_t(a,a')\|_{L^p}.$$

The noise term is bounded by (2). For the exact-gradient term, using (1) and Minkowski's inequality,

$$\begin{aligned}
\|\delta_{t+1}^{\mathrm{ex}}(a,a')\|_{L^p} &\leq \frac{1}{4A} \sum_{b \in \mathcal{Y}} \left(\|\delta_t(a,b)^2\|_{L^p} + \|\delta_t(a',b)^2\|_{L^p}\right) \\
&= \frac{1}{4A} \sum_{b \in \mathcal{Y}} \left(\|\delta_t(a,b)\|_{L^{2p}}^2 + \|\delta_t(a',b)\|_{L^{2p}}^2\right) \\
&\leq \frac{1}{2} M_t(2p)^2.
\end{aligned}$$

Taking the maximum over a, a' proves the claim. $\square$

Theorem 1 (Log-log transient via moment doubling)**.** *Assume $M_0(p) \le 1$ for all $p \ge 2$. Let $H \ge 1$ and define*

$$p_t := 2^{H-t+1}, \qquad x_t := M_t(p_t), \qquad t = 0, 1, \ldots, H.$$

Set

$$\varepsilon_H := 3\sigma 2^{H/2}.$$

Suppose

$$\varepsilon_H \le \frac{1}{4} \tag{3}$$

and

$$H \ge 3 + \left\lceil \log_2 \left(\frac{\log\left(1/\sqrt{2\varepsilon_H}\right)}{\log(4/3)} \right) \right\rceil. \tag{4}$$

Then

$$M_H(2) \le 2\varepsilon_H^2 + 3\sqrt{2}\,\sigma.$$

Consequently, if additionally

$$2\varepsilon_H^2 \le \sigma,$$

then

$$M_H(2) \le (3\sqrt{2} + 1)\sigma.$$

Proof. Since $2p_{t+1} = p_t$, Lemma 1 gives

$$\begin{aligned} x_{t+1} &= M_{t+1}(p_{t+1}) \\ &\le \frac{1}{2} M_t(2p_{t+1})^2 + 3\sigma\sqrt{p_{t+1}} \\ &= \frac{1}{2} x_t^2 + 3\sigma\sqrt{p_{t+1}}. \end{aligned}$$

Because $p_{t+1} \le 2^H$ for all $t = 0, \ldots, H-1$, we have

$$x_{t+1} \le \frac{1}{2} x_t^2 + \varepsilon_H. \tag{5}$$

Since $x_0 = M_0(2^{H+1}) \le 1$, the first step gives

$$x_1 \le \frac{1}{2} + \varepsilon_H \le \frac{3}{4},$$

where we used $\varepsilon_H \le 1/4$.

Define the hitting time

$$\tau := \inf\left\{ t \ge 1 : x_t < \sqrt{2\varepsilon_H} \right\}.$$

Before the hitting time, $x_t \ge \sqrt{2\varepsilon_H}$, hence

$$\varepsilon_H \le \frac{1}{2} x_t^2.$$

Therefore, for $1 \le t < \tau$, (5) implies

$$x_{t+1} \le x_t^2.$$

By induction, before the hitting time,

$$x_t \le \left(\frac{3}{4}\right)^{2^{t-1}}, \qquad t \ge 1.$$

Thus, it is enough to have

$$\left(\frac{3}{4}\right)^{2^{t-1}} \le \sqrt{2\varepsilon_H}.$$

Equivalently,

$$2^{t-1} \ge \frac{\log\left(1/\sqrt{2\varepsilon_H}\right)}{\log(4/3)}.$$

By (4), this occurs by time $H-2$. Hence

$$\tau \le H-2.$$

At the hitting time,

$$x_\tau < \sqrt{2\varepsilon_H}.$$

Therefore

$$x_{\tau+1} \le \frac{1}{2}x_\tau^2 + \varepsilon_H < \frac{1}{2}(2\varepsilon_H) + \varepsilon_H = 2\varepsilon_H.$$

Moreover, if $x_t \le 2\varepsilon_H$, then

$$x_{t+1} \le \frac{1}{2}(2\varepsilon_H)^2 + \varepsilon_H = 2\varepsilon_H^2 + \varepsilon_H \le 2\varepsilon_H,$$

where the last inequality follows from $\varepsilon_H \le 1/2$. Hence

$$x_{H-1} \le 2\varepsilon_H.$$

Finally, applying Lemma 1 with $p = 2$ gives

$$M_H(2) \le \frac{1}{2}M_{H-1}(4)^2 + 3\sqrt{2}\,\sigma.$$

Since $p_{H-1} = 4$, we have

$$M_{H-1}(4) = x_{H-1} \le 2\varepsilon_H.$$

Therefore

$$M_H(2) \le \frac{1}{2}(2\varepsilon_H)^2 + 3\sqrt{2}\,\sigma = 2\varepsilon_H^2 + 3\sqrt{2}\,\sigma.$$

If $2\varepsilon_H^2 \le \sigma$, then

$$M_H(2) \le (3\sqrt{2}+1)\sigma.$$

□

Corollary 1 (Dimension-free log-log transient)**.** *There exist absolute constants $\bar\sigma > 0$, independent of $A = |\mathcal{Y}|$, such that for every $\sigma \in (0, \bar\sigma)$, if*

$$H = \left\lceil \log_2 \log \frac{1}{\sigma} \right\rceil + \mathcal{O}(1),$$

then empirical DPO-Mix-R satisfies

$$\max_{a,a' \in \mathcal{Y}} \sqrt{\mathbb{E}|\delta(a, a'; \theta^{(H)})|^2} \le (3\sqrt{2}+1)\sigma.$$

Proof. With the above choice of H,

$$2^H = \Theta\left(\log \frac{1}{\sigma}\right).$$

Therefore

$$\varepsilon_H = 3\sigma 2^{H/2} = \Theta\left(\sigma\sqrt{\log \frac{1}{\sigma}}\right).$$

Hence, for all sufficiently small σ, we have

$$\varepsilon_H \leq \frac{1}{4}.$$

Also,

$$2\varepsilon_H^2 = 18\sigma^2 2^H = O\left(\sigma^2 \log \frac{1}{\sigma}\right) \leq \sigma$$

for all sufficiently small σ.

It remains to check the hitting-time condition. Since

$$\log \frac{1}{\sqrt{2\varepsilon_H}} = \frac{1}{2}\log\frac{1}{\sigma} - \frac{1}{4}\log\log\frac{1}{\sigma} + O(1),$$

we have

$$\log_2\left(\frac{\log\left(1/\sqrt{2\varepsilon_H}\right)}{\log(4/3)}\right) = \log_2\log\frac{1}{\sigma} + O(1).$$

The claim follows from Theorem 1. $\square$

References

[1] Ruizhe Shi, Runlong Zhou, and Simon S. Du. The crucial role of samplers in online direct preference optimization. *International Conference on Learning Representations*, 2025.